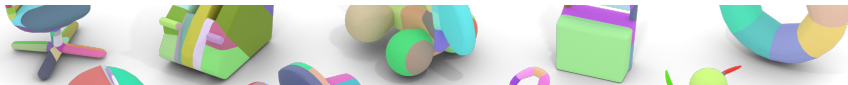
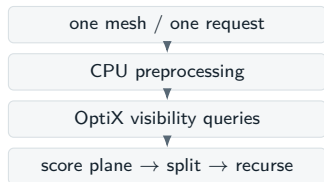


VisACD $\longrightarrow$ visacd_batched

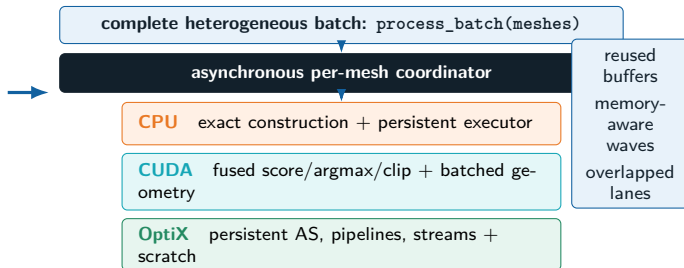
Same visibility-based decomposition; a new single-GPU throughput engine.



BASLINE VISACD



BATCHED EXECUTION ENGINE



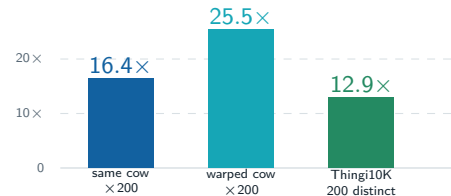
Throughput limit

Phase barriers, repeated setup and host-device traffic leave gaps between irregular jobs.

Same visibility objective and quality controls. **Deterministic batches keep the GPU fed.**

Three 200-mesh workloads: 12.9–25.5× faster

END-TO-END SPEEDUP VS. ORIGINAL PROCESS × 200



input set	original	batched	speedup
identical cow	57.8767 s	3.5276 s	16.4 ×
smooth variants	90.5219 s	3.5522 s	25.5 ×
Thing10K	130.3480 s	10.1113 s	12.9 ×

Diversity ladder 1 geometry hash → 200 smooth unique hashes (1.02% median RMS warp) → 11 categories, 116–48,023 vertices

Exactness: 33/33 surface + 44/44 full-remesh cases; scheduling matrices matched; Compute Sanitizer: 0 errors/hazards.

Medians of 3; concavity 0.04; 2 parts; conversion excluded. RTX 5090, CUDA 12.8, Threadripper PRO 9955WX. Original 3d9327b; current 01e398c; Thing10K manifest efbf9ed9b385cf9f. Pills: replicated-cow diagnostics.

WHAT MOVED THE NEEDLE

1 Overlap + persistence

per-mesh state machines; reused streams, buffers and OptiX state

2 Fuse + batch GPU work

scoring, deterministic argmax, clip prep and geometry stages

3 Move less data

compact crossing records, resident meshes and NumPy buffers

+24.7% surface preprocessing @ $n = 200$

91% less clip D→H traffic

Dense-resident exact preprocessing on Thing10K: 10.2873 s → 10.0478 s

(+2.3%; 7-run medians), same digest. The timing ranges overlap, so it remains opt-in.